PARENT ONLY — SOLVED ANSWER KEY

Vyom · Parent answer key

21 Dec – 27 Dec 2026 · Monday–Sunday · NDPS Class 9

Vyom ko test se pehle ye PDF mat dena. Student pack mein sirf questions aur blank answer space hai. Is alag parent PDF mein har original question ke saath uska matching solved answer, working aur marking points diye hain; Test ID se student paper match karein.

Checking: same evening. C = concept, R = recall, K = careless, T = time.

90%+: short revision. 75–89%: weak-topic practice + retest. Below 75%: topic dobara samjhao; mistakes notebook mein likho.

Answer keys existing source question banks se li gayi hain; school pattern abhi unconfirmed hai. Alternate valid answers ko marking guidance ke mutabik accept karein.

Week score register

Date / test ID	Test	Marks	Galti / retest
21 Dec D2026-12-21	Daily mixed Daily test	___ / 10	Retest: 24 Dec
22 Dec D2026-12-22	Daily mixed Daily test	___ / 10	Retest: 25 Dec
23 Dec D2026-12-23	Daily mixed Daily test	___ / 10	Retest: 26 Dec
24 Dec D2026-12-24	Daily mixed Daily test	___ / 10	Retest: 27 Dec
26 Dec D2026-12-26	Daily mixed Daily test	___ / 10	Retest: 29 Dec
27 Dec T2026-12-27	Maths Weekly test	___ / 40	Retest: 30 Dec

Daily Test · Answer key

Monday, 21 December 2026 · 10 marks

Test ID: D2026-12-21

Science Ch 8 Atom · Maths Ch 8 Sequences & Progressions

Q1 [1]

Question: Who discovered the electron?

Expected answer / marking points: J. J. Thomson.

Q2 [2]

Question: An atom has 11 protons and 12 neutrons. Find its atomic number, mass number and electronic configuration.

Expected answer / marking points: Z = 11, A = 23 [1]; 2, 8, 1 [1].

Q3 [2]

Question: Find the 15th term of the AP 20, 17, 14, ...

Expected answer / marking points: a = 20, d = –3 [1]; t_{15} = 20 + 14(–3) = –22 [1].

Q4 [2]

Question: The 4th term of an AP is 13 and the 9th term is 33. Find a and d.

Expected answer / marking points: 5d = 20 ⇒ d = 4 [1]; a = 13 – 12 = 1 [1].

Q5 [3]

Question: How many two-digit numbers are divisible by 3?

Expected answer / marking points: AP 12, 15, ..., 99 [1]; 99 = 12 + (n – 1)3 ⇒ n – 1 = 29 [1] ⇒ 30 numbers [1].

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Tuesday, 22 December 2026 · 10 marks

Test ID: D2026-12-22

SST Ch 1 · English Grammar + Writing · Science Ch 11 Reproduction · Maths Ch 14 Surface Area & Volume

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Q1 [1]
Question: Name the four disciplines of Social Science in your Grade 9 textbook.

Expected answer / marking points: History, Geography, Political Science, Economics.

Q2 [1]
Question: Give one reason why studying Social Science helps a citizen.

Expected answer / marking points: Understand how government/laws work; participate responsibly; ask informed questions; respect diversity (any one).

Q3 [1]
Question: Name the four disciplines of Social Science studied in Grade 9.

Expected answer / marking points: History, Geography, Political Science, Economics.

Q4 [1]
Question: Fill in the blank: Neither the teacher nor the students __ present. (was / were)

Expected answer / marking points: were.

Q5 [1]
Question: Use the correct modal: You __ wear a helmet while riding. It is the law. (must / may)

Expected answer / marking points: must.

Q6 [1]
Question: Change into reported speech: Ravi said, "I am tired."

Expected answer / marking points: Ravi said that he was tired.

Q7 [1]
Question: Name the type of asexual reproduction seen in Hydra.

Expected answer / marking points: Budding.

Q8 [1]
Question: What is pollination?

Expected answer / marking points: Transfer of pollen grains from the anther to the stigma.

Q9 [1]
Question: Write the formula for the volume of a cone.

Expected answer / marking points: $\frac{1}{3}\pi r^2 h$.

Q10 [1]
Question: Find the volume of a cube of edge 6 cm.

Expected answer / marking points: 216 cm^3.

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Wednesday, 23 December 2026 · 10 marks

Test ID: D2026-12-23

Hindi पाठ 1 · Maths Ch 8 Sequences & Progressions · SST P2 Ch 3 Resistance & Resilience · Maths Ch 14 Surface Area & Volume

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Q1 [1]

Question: हीरा और मोती का मालिक कौन था?

Expected answer / marking points: झूरी।

Q2 [2]

Question: कहानी के आधार पर हीरा और मोती की गहरी मित्रता के दो उदाहरण लिखिए।

Expected answer / marking points: मूक-भाषा में बातचीत; एक-दूसरे को चाटकर प्रेम जताना; संकट में साथ न छोड़ना (कोई दो) [1 + 1]।

Q3 [2]

Question: Find the sum 1 + 2 + 3 + ... + 50 using the triangular number formula.

Expected answer / marking points: $n(n + 1)/2 = 50 \times 51/2$ [1] = 1275 [1].

Q4 [2]

Question: Write a recursive formula for the GP 5, 15, 45, ...

Expected answer / marking points: $t_1 = 5$ [1]; $t_n = 3 \times t_{n-1}$ for $n \geq 2$ [1].

Q5 [1]

Question: In which year was the Vijayanagara Empire founded?

Expected answer / marking points: 1336 CE.

Q6 [1]

Question: Who won the First Battle of Panipat (1526)?

Expected answer / marking points: Babur.

Q7 [1]

Question: Write the formula for the curved surface area of a hemisphere.

Expected answer / marking points: $2\pi r^2$.

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Thursday, 24 December 2026 · 10 marks

Test ID: D2026-12-24

Science Ch 8 Atom · Hindi व्याकरण + लेखन · English Unit 6 · Science Ch 11 Reproduction

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Q1 [1]

Question: Who discovered the electron?

Expected answer / marking points: J. J. Thomson.

Q2 [2]

Question: An atom has 11 protons and 12 neutrons. Find its atomic number, mass number and electronic configuration.

Expected answer / marking points: Z = 11, A = 23 [1]; 2, 8, 1 [1].

Q3 [2]

Question: निम्नलिखित शब्दों में उपसर्ग लगाकर नए शब्द बनाइए: (i) मान (ii) योग

Expected answer / marking points: (i) अपमान / सम्मान [1] (ii) प्रयोग / सहयोग / वियोग [1]।

Q4 [2]

Question: वाच्य बदलिए: (i) राम पत्र लिखता है। (ii) मुझसे चला नहीं जाता।

Expected answer / marking points: (i) राम द्वारा पत्र लिखा जाता है (कर्मवाच्य) [1] (ii) मैं चल नहीं पाता (कर्तृवाच्य) [1]।

Q5 [2]

Question: What was Nabin’s rule about performances and why?

Expected answer / marking points: Only one performance every six months [1]; he felt that “familiarity breeds contempt” and an artist should work on the art [1].

Q6 [1]

Question: Which cell division produces gametes?

Expected answer / marking points: Meiosis.

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Saturday, 26 December 2026 · 10 marks

Test ID: D2026-12-26

English Unit 5 · AI Data Literacy · Hindi पाठ 7 मैं और मेरा देश · Maths Ch 14 Surface Area & Volume

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Q1 [2]

Question: What two choices did Deepa Malik have after becoming paralysed?

Expected answer / marking points: To waste her life in regret [1] or to turn it into a world of limitless possibilities — she chose sports [1].

Q2 [2]

Question: In “Nine Gold Medals”, why did all nine runners get gold medals?

Expected answer / marking points: When the smallest runner fell, the other eight went back to help him [1]; they finished together holding hands, so all were winners [1].

Q3 [1]

Question: What is the full form of DIKW?

Expected answer / marking points: Data, Information, Knowledge, Wisdom.

Q4 [2]

Question: Differentiate between data and information.

Expected answer / marking points: Data: raw facts/figures without context (e.g., 45, 60) [1]; information: processed data with meaning (e.g., “Vyom scored 45/60”) [1].

Q5 [2]

Question: लेखक के अनुसार व्यक्ति की पूर्णता किस पर निर्भर है?

Expected answer / marking points: केवल अपने घर-नगर पर नहीं [1]; अपने परिवार, क्षेत्र और राष्ट्र की पहचान और सम्मान से जुड़ी होती है [1]।

Q6 [1]

Question: How many litres is 1 m^3?

Expected answer / marking points: 1000 L.

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Maths · Weekly Test 2 · Answer key

Sunday, 27 December 2026 · 40 marks

Test ID: T2026-12-27

Class 9 · Ch 13 Two Variables, One Line · Ch 8 Sequences & Progressions · Ch 1 Coordinates · Ch 14 (shuruaat) + mixed Ch 4

Q1 [1]

Question: Which of the following is a solution of 2x + y = 7?

(a) (3, 2) (b) (1, 4)

(c) (2, 3) (d) (0, 6)

Expected answer / marking points: (c) (2, 3)

Q2 [1]

Question: The graph of y = 3 is a line parallel to:

(a) the x-axis (b) the y-axis

(c) the line y = x (d) none of these

Expected answer / marking points: (a) the x-axis

Q3 [1]

Question: The pair of equations x + 2y = 4 and 2x + 4y = 10 has:

(a) a unique solution (b) no solution

(c) infinitely many solutions (d) exactly two solutions

Expected answer / marking points: (b) no solution — 1/2 = 2/4 ≠ 4/10

Q4 [1]

Question: The next term of the GP 2, 6, 18, 54, ... is:

(a) 108 (b) 162

(c) 72 (d) 216

Expected answer / marking points: (b) 162

Q5 [1]

Question: The 10th triangular number is:

(a) 45 (b) 50

(c) 55 (d) 100

Expected answer / marking points: (c) 55 — 10 × 11 ÷ 2

Q6 [1]

Question: In an AP, $t_1 = 5$ and $d = -3$. Then t_6 =

(a) –10 (b) –13

(c) 20 (d) –8

Expected answer / marking points: (a) –10 — 5 + 5(–3)

Q7 [1]

Question: A point on the y-axis, 4 units below the origin, is:

(a) (4, 0) (b) (–4, 0)

(c) (0, –4) (d) (0, 4)

Expected answer / marking points: (c) (0, –4)

Q8 [1] **Question:** The volume of a cuboid measuring 10 cm × 6 cm × 4 cm is:

(a) 240 cm^3 (b) 120 cm^3

(c) 248 cm^3 (d) 20 cm^3

Expected answer / marking points: (a) 240 cm^3

Q9 [1] **Question:** Written as y = mx + d, the equation 3x + y = 6 has slope:

(a) 3 (b) –3

(c) 6 (d) –6

Expected answer / marking points: (b) –3 — y = –3x + 6

Q10 [1] **Question: A:** The equation 2x + 3y = 6 has infinitely many solutions.
R: A linear equation in two variables has infinitely many solutions.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (a)

Q11 [1] **Question: A:** 1, 4, 9, 16, ... is an AP.
R: In an AP, the difference between consecutive terms is constant.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (d) — differences 3, 5, 7 are not equal.

Q12 [2] **Question:** Find any three solutions of x – 2y = 4. Is (2, –1) a solution?

Expected answer / marking points: e.g., (4, 0), (0, –2), (6, 1) [1]. 2 – 2(–1) = 4 ✓, so yes [1].

Q13 [2] **Question:** For the sequence 7, 11, 15, 19, ..., write (i) an explicit formula (ii) a recursive formula.

Expected answer / marking points: (i) $t_n = 4n + 3$ [1]. (ii) $t_1 = 7$, $t_n = t_{n-1} + 4$ for $n \geq 2$ [1].

Q14 [2] **Question:** Find the total surface area of a cuboid 8 cm × 5 cm × 3 cm.

Expected answer / marking points: $2(8 \times 5 + 5 \times 3 + 3 \times 8) = 2(40 + 15 + 24)$ [1] = 158 cm^2 [1].

Q15 [2] **Question:** Expand (i) $(x - 2y)^2$ (ii) (3a + 1)(3a – 1).

Expected answer / marking points: (i) $x^2 - 4xy + 4y^2$ [1]. (ii) $9a^2 - 1$ [1].

Q16 [3] **Question:** Solve by substitution: 3x – y = 3 and 7x + 2y = 20.

Expected answer / marking points: y = 3x – 3 [1]; 7x + 2(3x – 3) = 20 ⇒ 13x = 26 ⇒ x = 2 [1]; y = 3 [0.5]; check 14 + 6 = 20 ✓ [0.5].

Q17 [3] **Question:** Which term of the AP 3, 8, 13, 18, ... is 248?

Expected answer / marking points: a = 3, d = 5 [1]; 3 + (n – 1)5 = 248 ⇒ n – 1 = 49 [1] ⇒ n = 50, the 50th term [1].

Q18 [3] **Question:** A GP has first term 3 and common ratio 2. Find its 8th term. Which term of this GP is 768?

Expected answer / marking points: $t_8 = 3 \times 2^7 = 384$ [1.5]; $3 \times 2^{n-1} = 768 \Rightarrow 2^{n-1} = 256 = 2^8 \Rightarrow n = 9$ [1.5].

Q19 [3] **Question:** Plot A(1, 1), B(5, 1) and C(5, 4). Find AB and BC, and then find AC. What type of triangle is ABC?

Expected answer / marking points: AB = 4 (same y), BC = 3 (same x) [1]; AB ⊥ BC, so AC = $\sqrt{(4^2 + 3^2)} = 5$ [1.5]; right-angled triangle (right angle at B) [0.5].

Q20 [4] **Question:**

> An auto-rickshaw charges a fixed ₹30 plus ₹12 for every kilometre travelled.

(i) Write the fare y (in ₹) for x km as a linear equation. [1]
(ii) Find the fare for 8 km. [1]
(iii) What do the slope and the y-intercept of this line mean? If a ride cost ₹174, how many km was it? [2]

Expected answer / marking points: (i) y = 12x + 30 [1]. (ii) 12 × 8 + 30 = ₹126 [1]. (iii) Slope 12 = cost per km; y-intercept 30 = fixed charge [1]; 12x + 30 = 174 ⇒ x = 12 km [1].

Q21 [5] **Question:** Draw the graphs of x + y = 6 and x – y = 2 on the same axes. Find (i) their point of intersection (ii) the points where they meet the x-axis (iii) the area of the triangle formed by the two lines and the x-axis.

ATHVA / OR

A closed cylindrical water tank has radius 1.4 m and height 2 m. Find (i) its volume in m^3 and in litres (ii) its curved surface area (iii) its total surface area.

Expected answer / marking points: Tables + correct graph [2]. (i) (4, 2) [1]. (ii) (6, 0) and (2, 0) [1]. (iii) Base = 6 – 2 = 4, height = 2 ⇒ area = ½ × 4 × 2 = 4 sq units [1].
OR: (i) $\pi r^2 h = 22/7 \times 1.96 \times 2 = 12.32\ m^3 = 12{,}320$ L [2]. (ii) $2\pi rh = 2 \times 22/7 \times 1.4 \times 2 = 17.6\ m^2$ [1.5]. (iii) $2\pi r(h + r) = 2 \times 22/7 \times 1.4 \times 3.4 = 29.92\ m^2$ [1.5].

Score: ___ / 40 Checked by: _____

Weak topic aur correction:

Parent Sunday handover checklist

- [ ] Agle Monday–Sunday ka student pack print karke diya.
- [ ] Parent solved key alag rakhi; student copy ke saath share nahi ki.
- [ ] Pichhle week ka pending kaam aur 3 weak topics review kiye.
- [ ] School exam date sheet aur tuition timings se schedule match kiya.

Agli priority / teacher se poochhna: